MARKET OPPORTUNITY ANALYSIS AGENT

Beginner-friendly implementation guide for your team project

Your role	Member 1 — Market Opportunity Analysis Agent
Main goal	Analyze a startup idea and produce a structured report about its market, demand, growth, and potential size.
Suggested starter stack	Python for the agent logic; a simple command-line interface first. Add an AI API and web search after the basic flow works.

What you will build

The user enters a startup idea and target market. Your agent organizes the analysis into clear sections: industry size and trends, customer demand, growth opportunities, and TAM/SAM/SOM estimates when trustworthy data is available. It must label assumptions and avoid making up statistics.

Important: TAM, SAM, and SOM should be estimates supported by sources or transparent assumptions. If reliable data is missing, write “insufficient data” instead of inventing a number.

1. Understand the key terms

Term	Simple meaning	Example for a food-delivery startup
TAM	Total Addressable Market: the whole market if you could serve everyone relevant.	All potential food-delivery spending in the defined country/market.
SAM	Serviceable Available Market: the part of TAM your product and geography can realistically serve.	Food-delivery customers in the cities you plan to launch in.
SOM	Serviceable Obtainable Market: the realistic share you may capture in the near term.	A cautious estimate of customers or revenue you might win in the first few years.

2. Agent workflow

1. Collect the startup idea, target customer, geography, and business model.

2. Identify what market information is needed (industry size, growth, trends, demand, pricing, regulations).

3. Search trustworthy sources such as government statistics, industry reports, company filings, and reputable research publications.

4. Extract facts and record each source URL, publication date, and the exact claim it supports.

5. Estimate TAM/SAM/SOM only when the assumptions and inputs are visible.

6. Generate a structured report with findings, opportunities, risks, data gaps, and sources.

7. Pass the report to the team's main agent or combine it with customer and competitor agents.

3. Suggested project folder structure

```
market-opportunity-agent/
■■■ main.py              # Starts the agent
■■■ market_agent.py      # Main analysis logic
■■■ prompts.py           # Instructions for the AI model
■■■ sources.py           # Search/source handling (add later)
■■■ requirements.txt     # Python packages
■■■ README.md            # Setup and run instructions
```

4. First working version (no API required)

Start with a simple input/output prototype. This confirms your workflow before you add an AI API or web search. Save the following as **main.py** and run it with **python main.py**.

```
startup_idea = input("Enter your startup idea: ").strip()
target_market = input("Enter target location/customer: ").strip()
business_model = input("How will it make money? ").strip()

if not startup_idea or not target_market:
    print("Please enter both a startup idea and target market.")
else:
    print("\\n=== MARKET OPPORTUNITY ANALYSIS ===")
    print("Startup idea:", startup_idea)
    print("Target market:", target_market)
    print("Business model:", business_model)

    print("\\n1. Industry size and trends")
    print("Research market size, growth rate, and major trends.")
    print("Do not add numbers until you have a reliable source.")

    print("\\n2. Market demand")
    print("List evidence of customer demand and the problem being solved.")

    print("\\n3. Growth opportunities")
    print("Identify underserved areas, new customer groups, or useful features.")

    print("\\n4. TAM / SAM / SOM")
    print("TAM: [source-backed estimate or insufficient data]")
    print("SAM: [explain geography/product filters]")
```

```
    print("SOM: [show assumptions and calculation]")

    print("\\n5. Risks and data gaps")
    print("List competition, regulation, cost, and missing evidence.")

    print("\\nNext step: add AI generation and source-backed web research.")
```

This is a prototype, not the final AI agent: it prints a report outline rather than doing real research. The next step is to connect an AI model and a web-search tool, then require citations for factual claims.

5. Prompt for the AI agent

```
You are a Market Opportunity Analysis Agent. Analyze the startup idea and target market provided by the user.

Return these sections:
1. Startup idea and scope
2. Industry size and recent market trends
3. Market demand and evidence
4. Growth opportunities
5. TAM, SAM, and SOM (only estimate when there is enough evidence; show formula, inputs, assumptions, and units)
6. Risks and uncertainties
7. Data gaps and recommended next research steps
8. Sources: source name, URL, publication date (if available), and which claim it supports

Rules:
- Never invent statistics, sources, or URLs.
- Separate verified facts from estimates and assumptions.
- Prefer recent, reputable, primary sources.
- If data is unavailable, say “insufficient reliable data”.
- Explain findings in simple language.
- State the geography, time period, and currency for every market-size estimate.
```

6. Output format to share with the other team members

Section	What it should contain
Executive summary	Short view of the opportunity and the biggest uncertainty.
Market trends	Trend, why it matters, evidence/source.
Demand	Customer problem, evidence of demand, limitations.
TAM/SAM/SOM	Value, currency, year, formula, assumptions, source.
Opportunities	Prioritized growth ideas with reasons.
Risks	Market, operational, regulatory, and competitive risks.
Sources	Clickable URLs and the claims each source supports.

7. How to finish and demonstrate your task

• Run the prototype and test it with two different startup ideas.

• Add an AI model call using the API your team has selected; keep API keys in environment variables, never commit them to GitHub.

• Add web search or a trusted data source and save source URLs with the findings.

• Check that each statistic has a source and that TAM/SAM/SOM calculations show assumptions.

• Export one sample analysis as a PDF or Markdown report.

• Share the function/interface with your teammates so the main project can call your agent.

8. What to tell your team

“I’m building the Market Opportunity Analysis Agent. It will take a startup idea and target market, organize research on industry trends and demand, estimate TAM/SAM/SOM only when evidence supports it, and return a structured report with sources, assumptions, risks, and data gaps. I’ll first build and test the basic flow, then integrate AI and web research.”

Note: Your screenshot shows only part of the team’s full instructions. Confirm the required programming language, AI model, and integration format with your team before final submission.